ARJUNA JEE 2027

Lecture -04 | Mathematics

TRIGONOMETRIC FUNCTIONS

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Topics *to be covered*

A **Definition of Sine & Cosine**

B **Reduction Formula**

C **Value at Larger Angles**

D **Question Practice**





LAST CLASS RECAP

Answer the following Questions:

1. Angles measured in anticlockwise direction are ____+ve____.
2. Angles measured in clockwise direction are ____-ve.____.
3. Relation between degree and radians is ____π rad. = 180°____.
4. 1° = ____60′____ minutes.
5. $1'$ (one minute) = ____60″____ seconds.
6. Approximate values of: π = ____3.14 or $\frac{22}{7}$____, $\frac{\pi}{2}$ = ____1.57____.

$\frac{\pi}{180}$ rad = 1°

1 rad = $\left(\frac{180}{\pi}\right)^\circ$

$\frac{22}{7} > \pi$



LAST CLASS RECAP

Answer the following Questions:

7. $(22.5)^\circ = 22^\circ 30'$ (True/False)

8. $\frac{7\pi}{8} =$ 157°30′ .

9. 3 radians $\cong 171^\circ 49' 60''$ (True/False)

10. $45^\circ 15' 30'' = \left(\frac{5431}{120}\right)^\circ = \left(\frac{5431}{120} \times \frac{\pi}{180}\right)^c$ (True/False)

$45^\circ + 15' + 30''$

$45^\circ + \left(15 \times \frac{1}{60}\right)^\circ + \left(30 \times \frac{1}{3600}\right)^\circ$

NCERT

(7) $(22.5)^\circ = 22^\circ + 0.5^\circ$

0.5° → minutes.

$= 22^\circ + 30' = 22^\circ 30'$

$1^\circ = 60'$

$0.5^\circ = (0.5 \times 60)'$

$= 30'$

(8) $\frac{7\pi}{8} = \frac{7\cancel{\pi}}{\cancel{8}_{2}} \times \frac{\overset{45}{\cancel{180}}}{\cancel{\pi}} = \left(\frac{315}{2}\right)^\circ = (157.5)^\circ = 157^\circ + 0.5^\circ$

$= 157^\circ 30'$

(9) $1 \text{ rad.} = \left(\frac{180}{\pi}\right)^\circ$

$3 \text{ rad.} = \left(\frac{180}{\frac{22}{7}} \times 3\right)^\circ = \left(\frac{\cancel{180}^{90}}{\cancel{22}_{11}} \times 21\right)^\circ = \left(\frac{1890}{11}\right)^\circ = (171.81)^\circ$

$= 171^\circ + (0.81)^\circ$

$= 171^\circ + 48' + 0.6'$

$= 171^\circ + 48' + 36''$

$= 171^\circ 48' 36''$

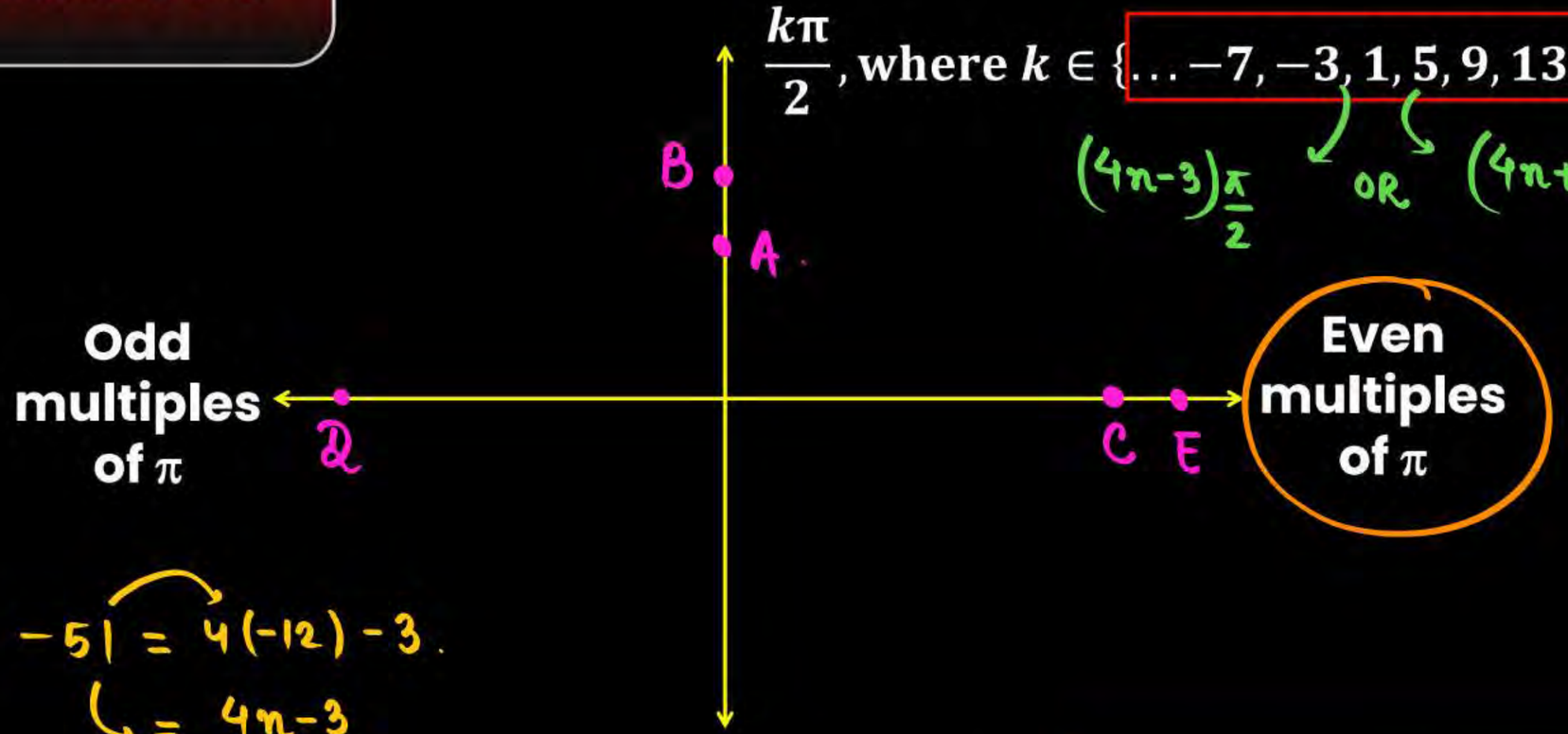
$1^\circ = 60'$

$0.81^\circ = \left(60 \times \frac{0.81}{100}\right)' = (48.6)'$

$1' = 60''$

$0.6' = (0.6 \times 60)'' = 36''$

QUADRANTS



(A) $-\frac{51\pi}{2}$

(B) $\frac{173\pi}{2}$

(C) 108π

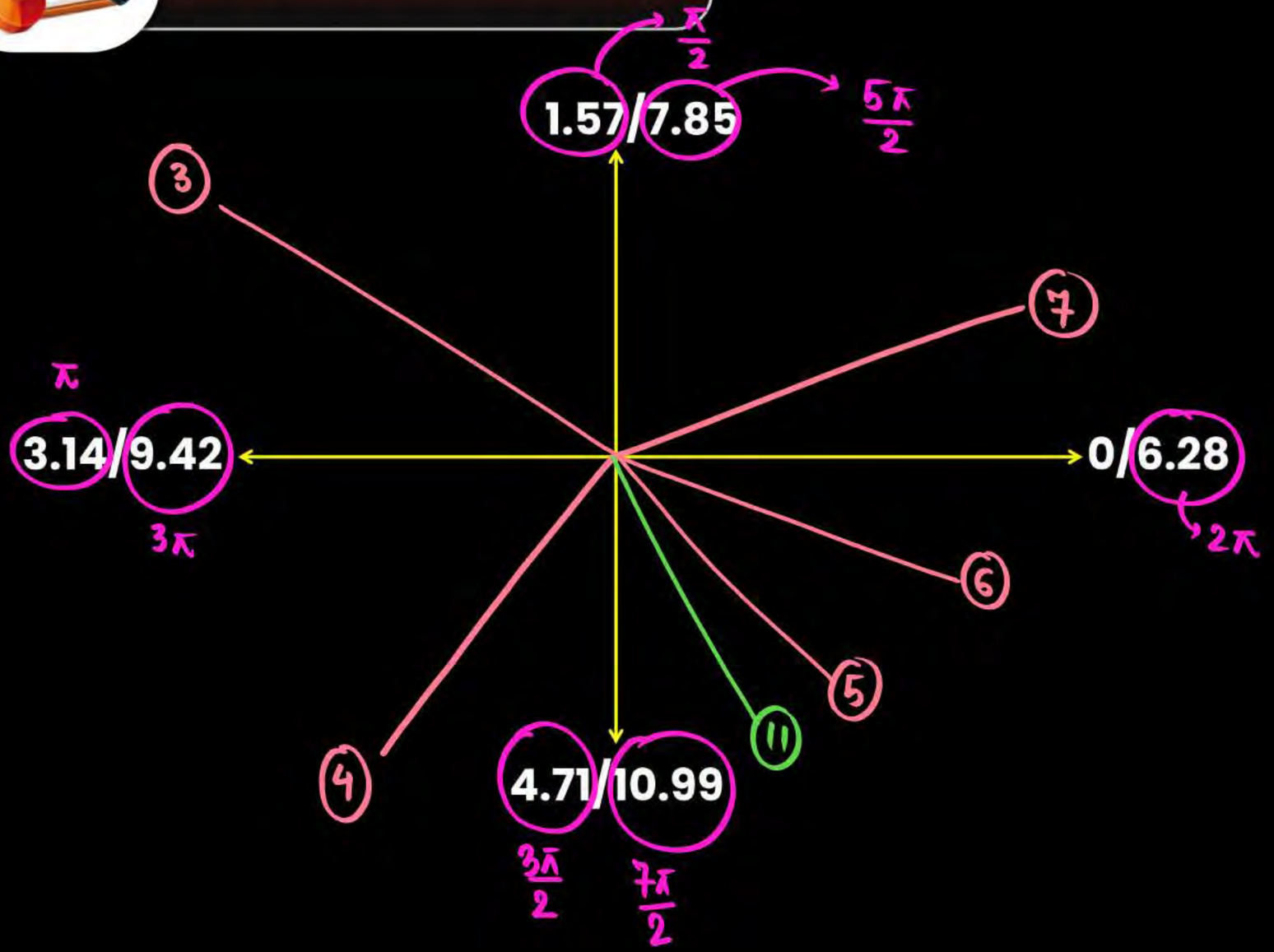
(D) 17345π

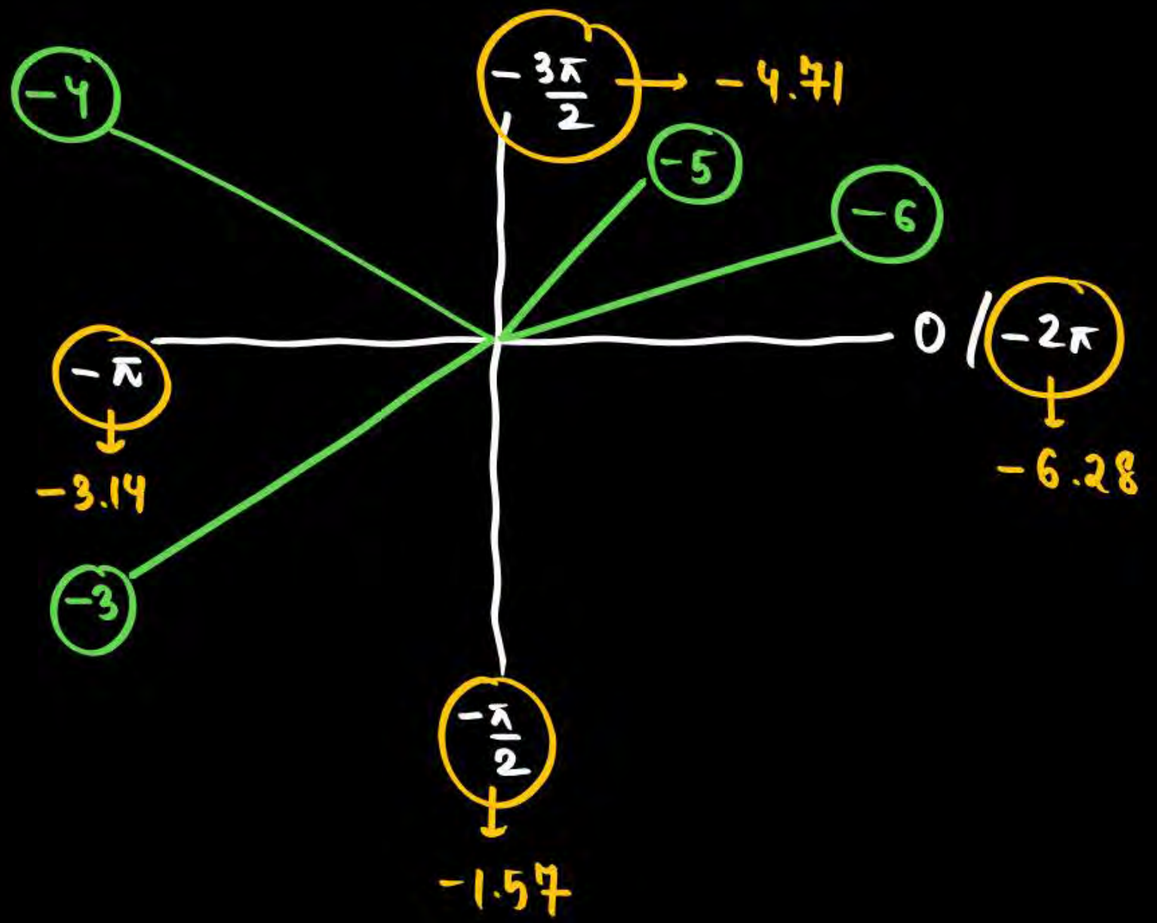
(E) -7134π

$-51 = 4(-12) - 3$

$= 4n - 3$

QUADRANTS




-4
$-\frac{3\pi}{2}$
-4.71
-5
-6
0
-2π
$-\pi$
-3.14
-6.28
-3
$-\frac{\pi}{2}$
-1.57

Question Practice

Question [JEE Mains-2021]

Same Ques^{n} as of last class.

If $15\sin^4\alpha + 10\cos^4\alpha = 6$, for some $\alpha \in R$, then the value of $27\sec^6\alpha + 8\operatorname{cosec}^6\alpha$ is equal to:

A 500

B 400

C 250

D 350

$\div 30$

$$\frac{\sin^4\alpha}{2} + \frac{\cos^4\alpha}{3} = \frac{1}{5}$$

Titu's lemma

$$\frac{\sin^2\alpha}{2} = \frac{\cos^2\alpha}{3}$$

$$\tan^2\alpha = \frac{2}{3}$$ ✓

RAPID FIRE ROUND

Identify Quadrants of following angles:

large angle ÷ by 90°

(i) $330°$ → IV

(ii) $-210°$ → II

(iii) $-1120° = -(90° \times 12 + 40°)$

$= -90° \times 12 - 40°$

$-180° \times 6 - 40°$ → IV

(iv) $\frac{11\pi}{5} = \frac{10\pi}{5} + \frac{\pi}{5} = 2\pi + \frac{\pi}{5}$ → I ($\frac{\pi}{5} = 36°$)

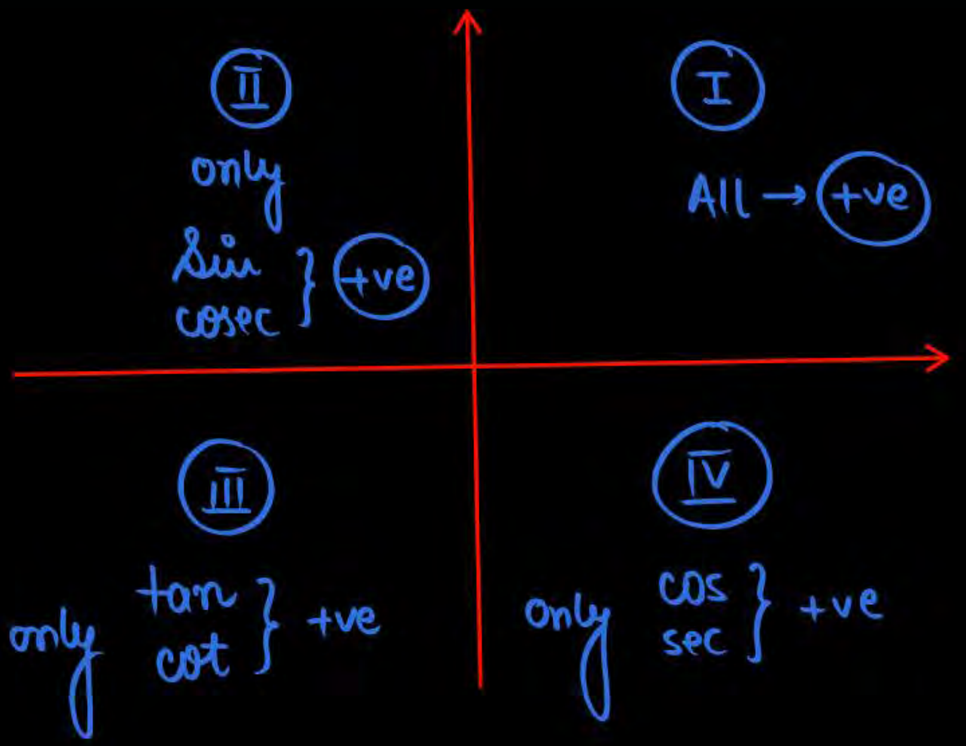
(v) $\frac{57\pi}{4} = \frac{56\pi}{4} + \frac{\pi}{4} = 14\pi + \frac{\pi}{4}$ → I

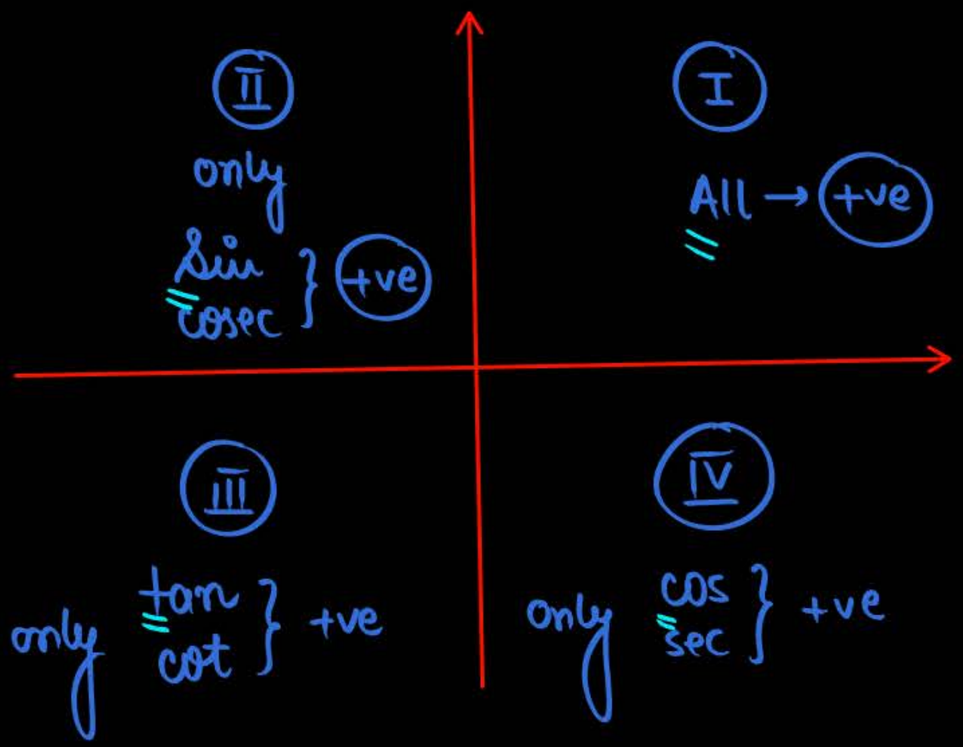
(vi) $\frac{-31\pi}{6} = -\left(\frac{30\pi}{6} + \frac{\pi}{6}\right) = -5\pi - \frac{\pi}{6}$ → II

(vii) 5

(viii) 11

SIGN OF TRIGONOMETRIC RATIO's





(1) $\sec(803^\circ) = \underline{+ve}$

I

803°

$90^\circ \times 8 + 83^\circ$

$180^\circ \times 4 + 83^\circ$

(2) $\sin(11) = \underline{-ve}$.

IV

(3) $\tan(4) = +ve$.

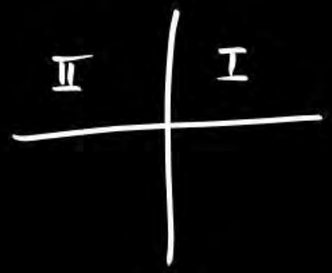
III

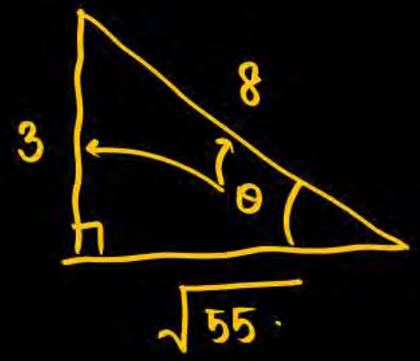
(4) $\operatorname{cosec}(-1120^\circ) = -ve$.

IV

+ve $\Rightarrow$ $\theta \in$ I or II

$\sin\theta = \frac{3}{8} = \frac{P}{H}$





If $\theta \in$ I

$$\cos\theta = +\frac{\sqrt{55}}{8}$$

If $\theta \in$ II

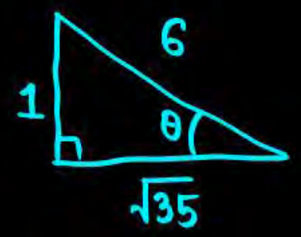
$$\cos\theta = -\frac{\sqrt{55}}{8}$$

$$\sin\theta = \frac{3}{8} \rightarrow \cos\theta = +\frac{\sqrt{55}}{8}, \quad \cos\theta = -\frac{\sqrt{55}}{8}$$

RAPID FIRE ROUND

III OR IV

$\sin\theta = -\frac{1}{6}$



Identify sign of following:

(i) If $\sin\theta = -\frac{1}{6}$ then $\cos\theta =$ $\frac{\mp\sqrt{35}}{6}$.

(ii) If $\sin\theta = -\frac{1}{6}$ & θ lies in IV quadrant then $\cos\theta =$ $\frac{+\sqrt{35}}{6}$.

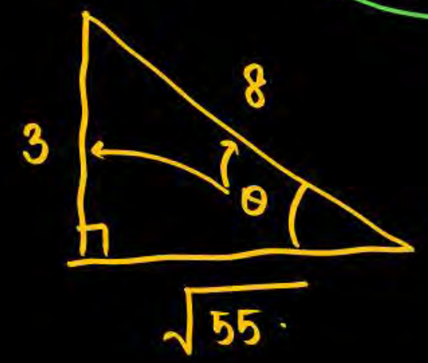
(iii) If $\sin\theta = -\frac{2}{5}$ & $\cot\theta = \frac{\sqrt{21}}{2}$ then $\sec\theta =$ $-\frac{5}{\sqrt{21}}$.

($\sin\theta$: θ → ⊖; $\cot\theta$: θ → ⊕)



$\theta \in$ III quad.

$\boxed{\sin\theta = \frac{3}{8}} = \frac{P}{H}$



$\cos\theta = \frac{\sqrt{55}}{8}$ ✗

Question

If θ is an acute angle such that $5\cot\theta = 3$, then find the value of $\frac{(5\sin\theta - 3\cos\theta)}{(4\sin\theta + 3\cos\theta)}$

(I)

$0 < \theta < 90^\circ$

A $\frac{11}{18}$

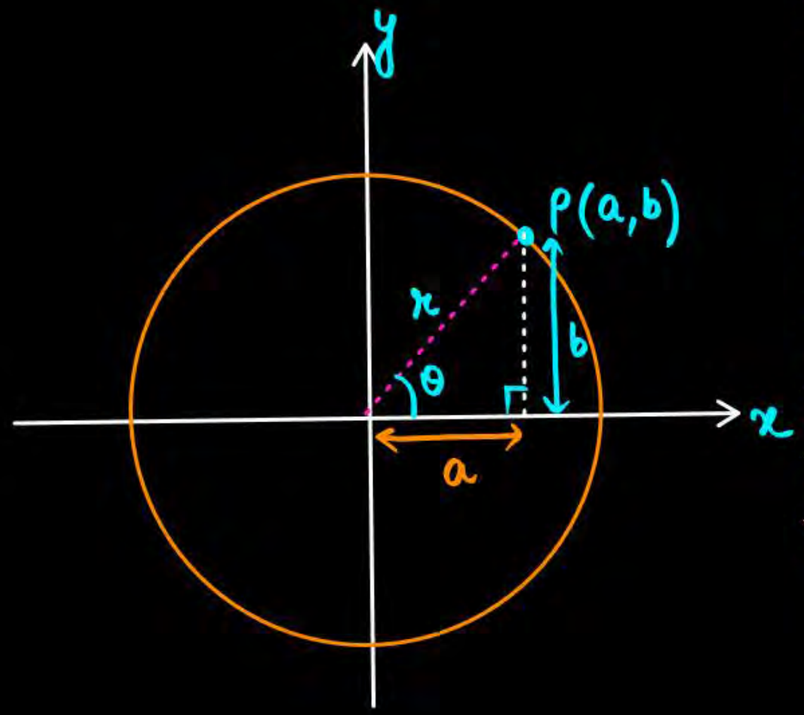
B $\frac{16}{29}$

C $\frac{14}{27}$

D None of these

DIBY — (A)

REAL DEFINITION OF SINE & COSINE

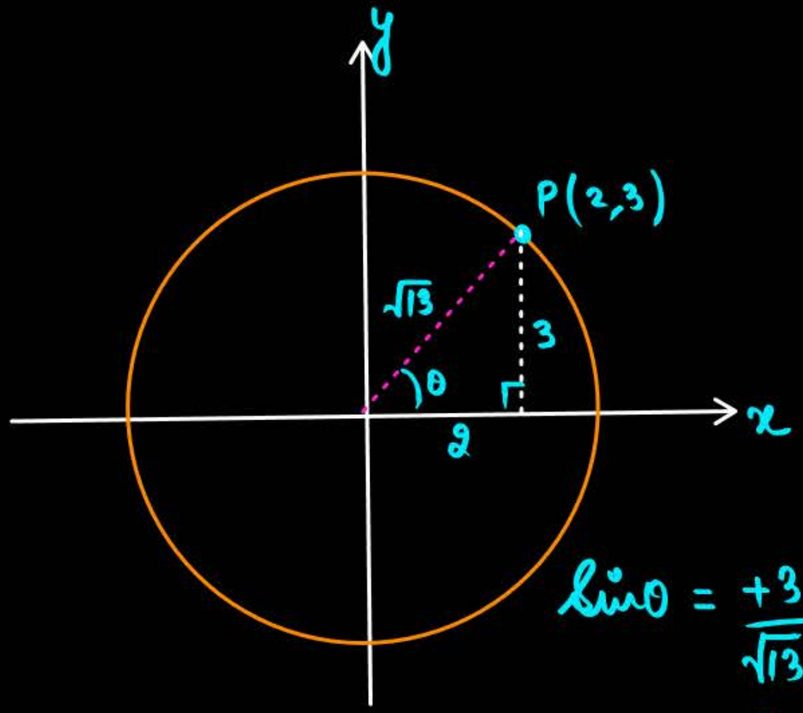


$$\sin\theta = \frac{y\text{-coord. of } P}{r}$$

$$\cos\theta = \frac{x\text{-coord. of } P}{r}$$

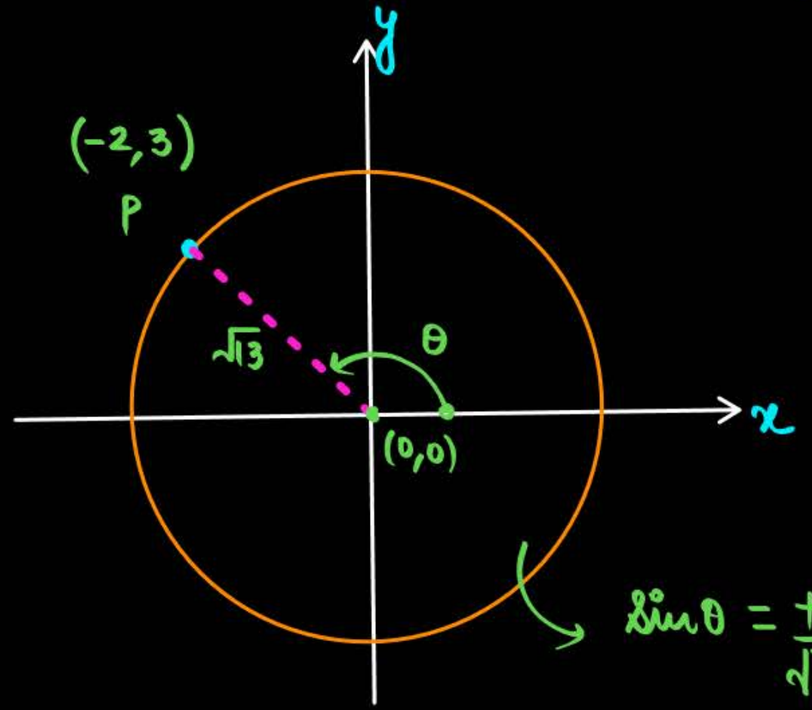
$$\sin\theta = \frac{y\text{-coord. of } P}{r}$$

$$\cos\theta = \frac{x\text{-coord. of } P}{r}$$

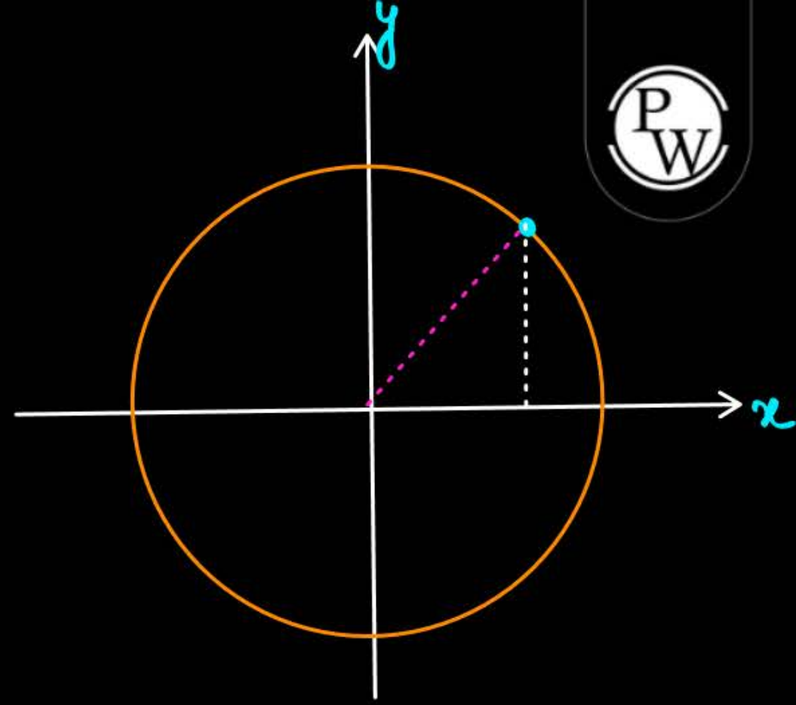


$$\sin\theta = \frac{+3}{\sqrt{13}}$$

$$\cos\theta = \frac{+2}{\sqrt{13}}$$

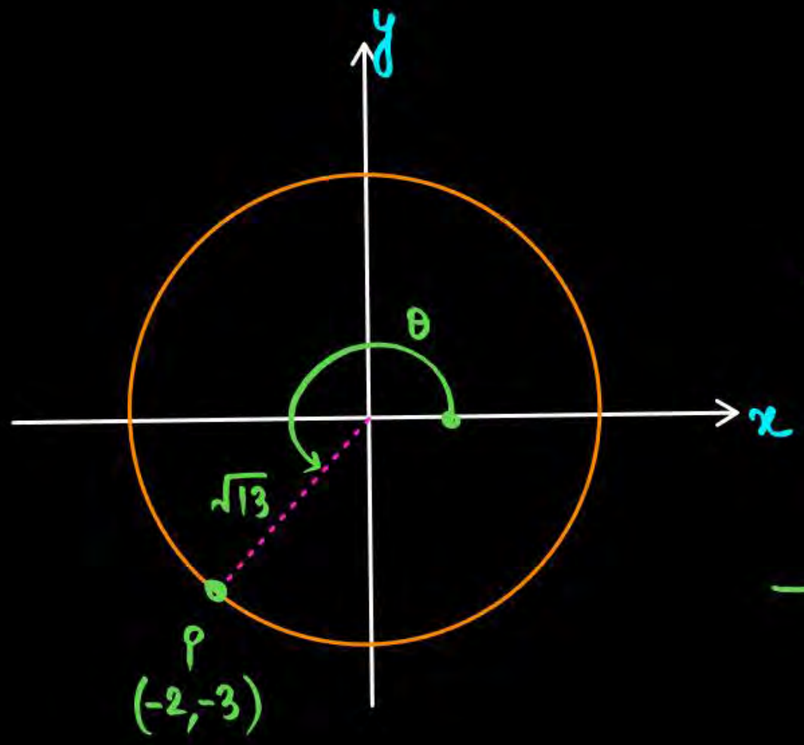


$$\sin\theta = \frac{+3}{\sqrt{13}} \qquad \cos\theta = \frac{-2}{\sqrt{13}}$$



$$\sin\theta = \frac{y\text{-coord. of } P}{r}$$

$$\cos\theta = \frac{x\text{-coord. of } P}{r}$$



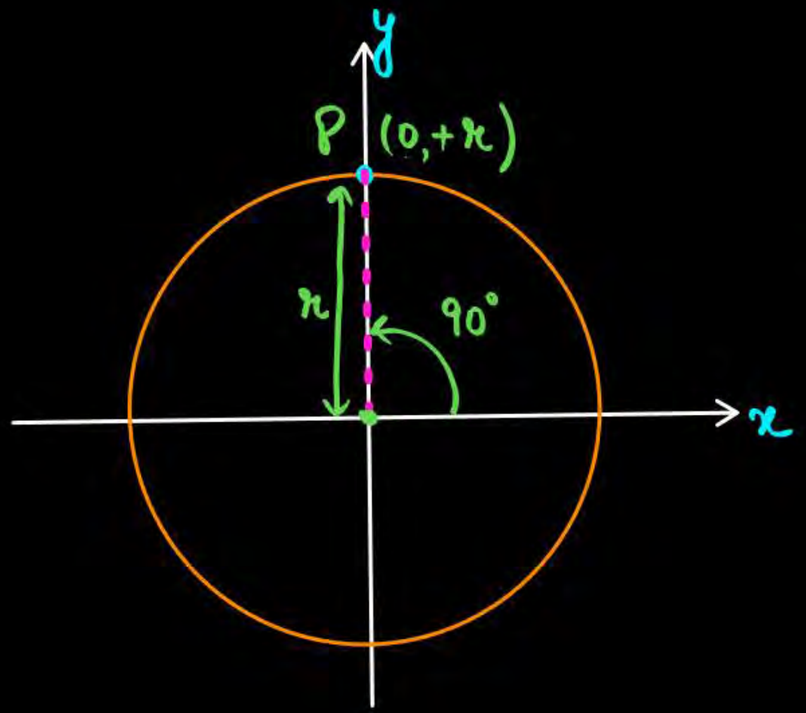
$$\sin\theta = -\frac{3}{\sqrt{13}}$$

$$\cos\theta = -\frac{2}{\sqrt{13}}$$

$$\tan\theta = +\frac{3}{2}$$

$$\sin\theta = \frac{y\text{-coord. of } P}{r}$$

$$\cos\theta = \frac{x\text{-coord. of } P}{r}$$

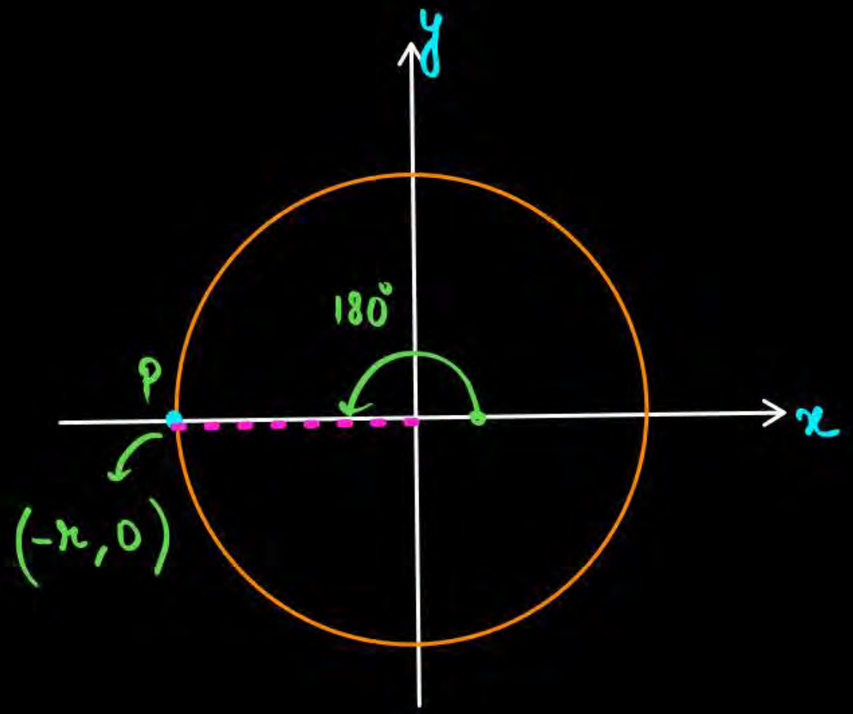


$$\sin 90^\circ = \frac{+r}{r} = 1.$$

$$\cos 90^\circ = \frac{0}{r} = 0.$$

$$\sin\theta = \frac{y\text{-coord. of } P}{r}$$

$$\cos\theta = \frac{x\text{-coord. of } P}{r}$$

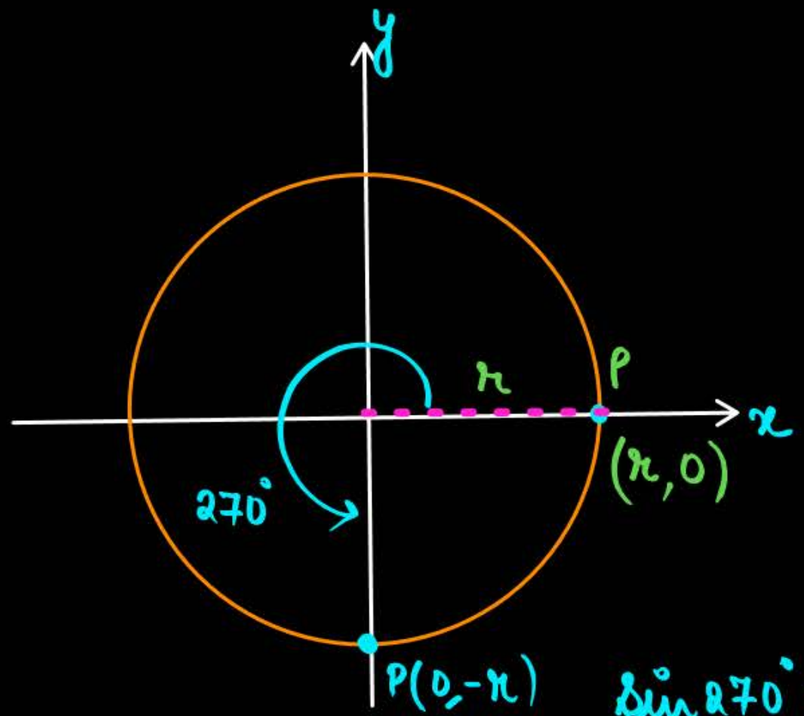


$$\sin 180^\circ = \frac{0}{r} = 0$$

$$\cos 180^\circ = \frac{-r}{r} = -1$$

$$\sin\theta = \frac{y\text{-coord. of } P}{r}$$

$$\cos\theta = \frac{x\text{-coord. of } P}{r}$$



$$\cos 0^\circ = \frac{+r}{r} = 1$$

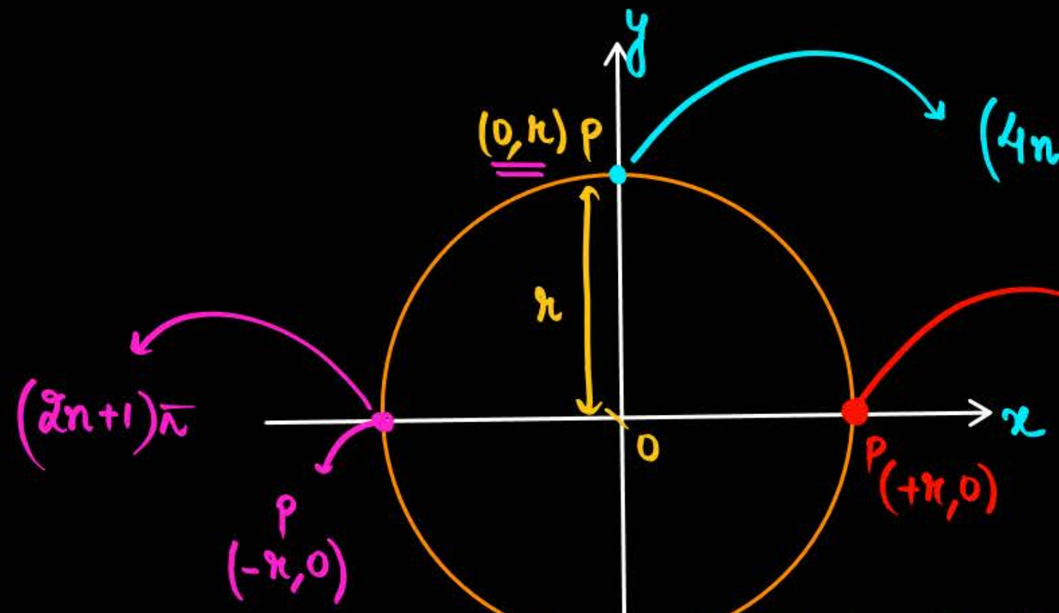
$$\sin 0^\circ = \frac{0}{r} = 0$$

$$\sin 270^\circ = \frac{-r}{r} = -1$$

$$\cos 270^\circ = \frac{0}{r} = 0$$

$$\sin\theta = \frac{y\text{-coord. of } P}{r}$$

$$\cos\theta = \frac{x\text{-coord. of } P}{r}$$



$\sin 2n\pi = 0$

$\cos 2n\pi = 1$.

$\sin\left((2n+1)\pi\right) = \frac{0}{r} = 0$

$\cos\left((2n+1)\pi\right) = \frac{-r}{r} = -1$

$P(0,-r)$

$\sin\left((4n-1)\frac{\pi}{2}\right) = -1$

$\cos\left((4n-1)\frac{\pi}{2}\right) = 0$.

$\sin\left((4n+1)\frac{\pi}{2}\right) = \frac{+r}{r} = 1$.

$\cos\left((4n+1)\frac{\pi}{2}\right) = \frac{0}{r} = 0$.

DEALING WITH NEGATIVE ANGLE

$$\cos(-\theta) = \cos\theta$$
$$\sec(-\theta) = \sec\theta$$

ex:- $\cos(-30^\circ) = \cos(30^\circ)$ ✓

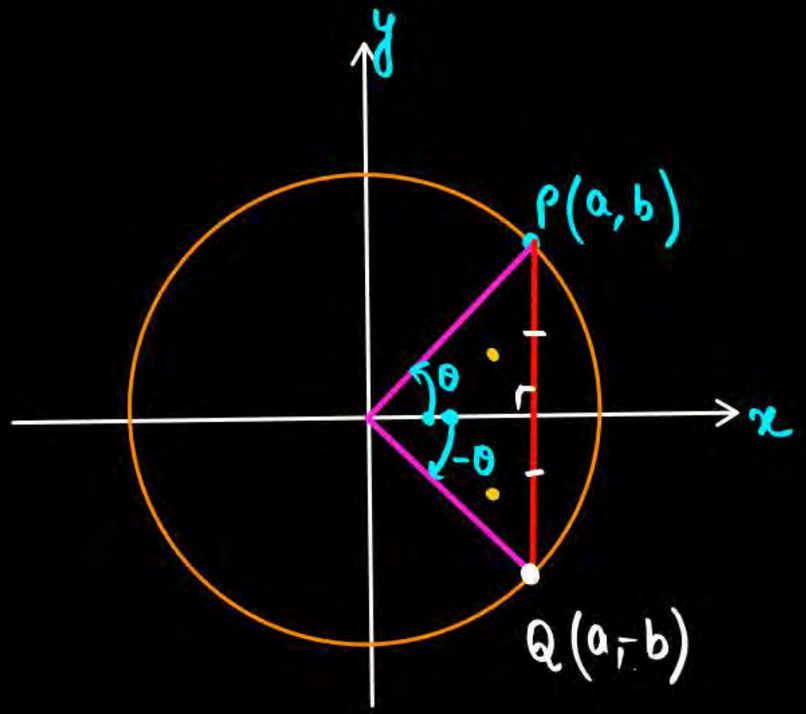
$\sec(-180^\circ) = \sec(180^\circ)$ ✓

$$\sin(-\theta) = -\sin\theta$$
$$\operatorname{cosec}(-\theta) = -\operatorname{cosec}\theta$$
$$\tan(-\theta) = -\tan\theta$$
$$\cot(-\theta) = -\cot\theta$$

ex:- $\sin(-60^\circ) = -\sin 60^\circ = -\frac{\sqrt{3}}{2}$

$$\sin\theta = \frac{y\text{-coord. of } P}{r}$$

$$\cos\theta = \frac{x\text{-coord. of } P}{r}$$



$$\sin\theta = \frac{\pm b}{r}$$

$$\cos\theta = \frac{\pm a}{r}$$

$$\sin(-\theta) = -\frac{b}{r} = -\sin\theta$$

$$\cos(-\theta) = \frac{+a}{r} = \cos\theta$$



REDUCTION FORMULA

angle

S-0: make angle ⊕

S-I: decide Quadrant & mark sign

S-II: $\text{angle} = a\left(\frac{\pi}{2}\right) \pm b$

- odd ⇒ sin ⟷ cos, tan ⟷ cot, sec ⟷ cosec
- even ⇒ no Change in trigono. func^{n}.

$\theta \rightarrow$ acute

(1)

odd

IV

$\sin\left(\frac{3\pi}{2} + \theta\right) = -\cos\theta$

$\sin\left(\frac{3\pi}{2} - \theta\right) = -\cos\theta$

III

III

$\sin(\pi + \theta) = \sin\left(2\cdot\frac{\pi}{2} + \theta\right) = -\sin\theta$ ✓

even

II

$\sin(\pi - \theta) = +\sin\left(2\cdot\frac{\pi}{2} - \theta\right) = +\sin\theta$ ✓

III

$\tan(\pi + \theta) = +\tan\theta$ ✓

$2\left(\frac{\pi}{2}\right) + \theta$

even

odd

III

$\tan\left(\frac{3\pi}{2} - \theta\right) = +\cot\theta$ ✓

II ← $\frac{\pi}{2} + \theta$

I ← $\frac{\pi}{2} - \theta$

III ← $\pi + \theta$

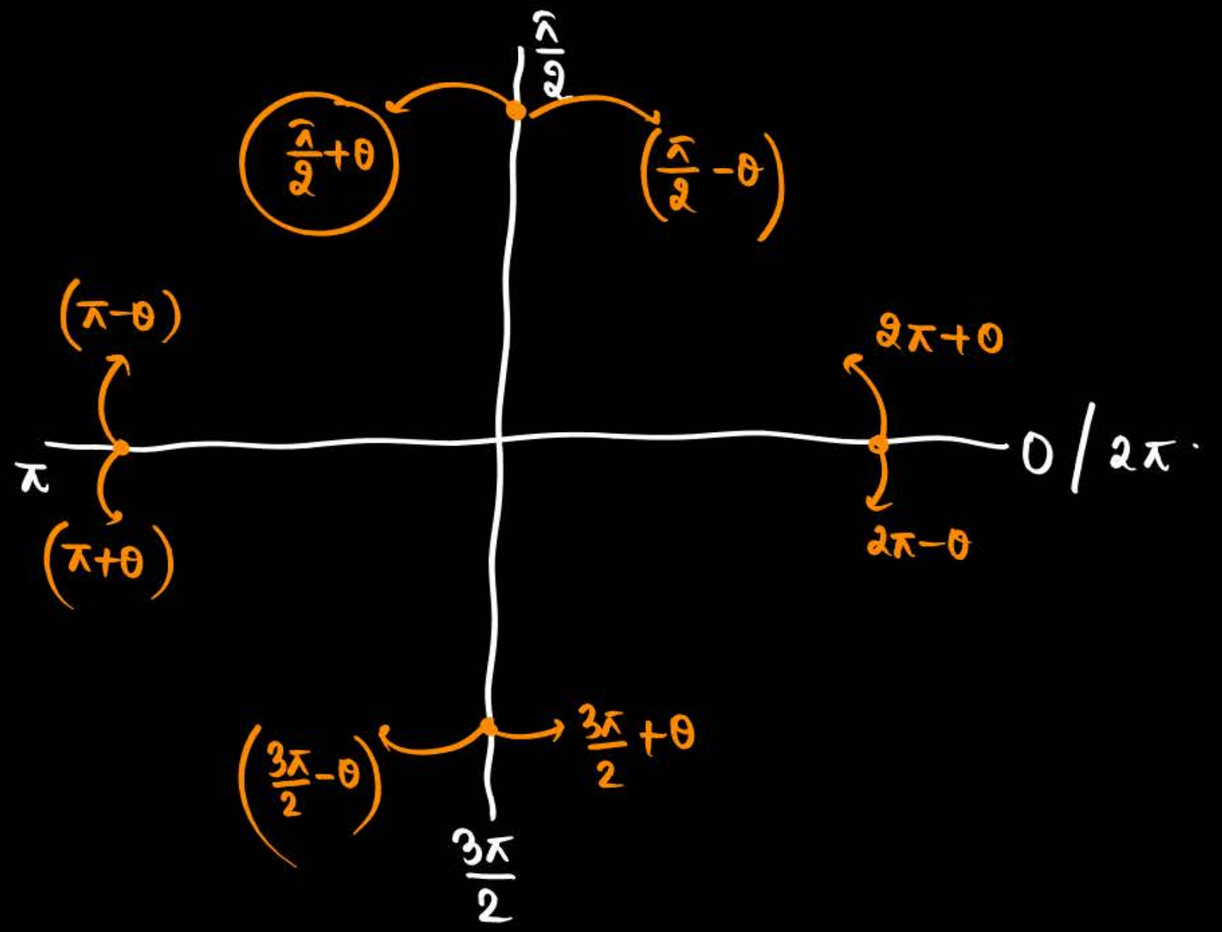
II ← $\pi - \theta$

IV ← $\frac{3\pi}{2} + \theta$

III ← $\frac{3\pi}{2} - \theta$

I ← $2\pi + \theta$

IV ← $2\pi - \theta$



DO IT BY YOURSELF (DIBY)

DIBY-01

If $\sin A \tan A = \cos^2 A$, then $\cos^3 A + \cos^2 A$ is equal to

(A) 1 (B) 2 (C) 4 (D) None of these

[Ans. : (A)]

DIBY-02

If $\text{cosec}A + \cot A = \frac{11}{2}$, then $\tan A$ is

(A) $\frac{21}{22}$ (B) $\frac{15}{16}$ (C) $\frac{44}{117}$ (D) $\frac{117}{43}$

[Ans. : (C)]

DIBY-03

If $\tan\theta = \frac{-4}{3}$, then $\sin\theta$ is

(A) $\frac{-4}{5}$ but not $\frac{4}{5}$ (B) $\frac{-4}{5}$ or $\frac{4}{5}$ (C) $\frac{4}{5}$ but not $-\frac{4}{5}$ (D) None of these

[Ans. : (B)]

DIBY-04

If $\tan\theta + \sin\theta = m$ and $\tan\theta - \sin\theta = n$, then prove that $m^2 - n^2 = 4\sin\theta\tan\theta$

DIBY-05

Prove that $\dfrac{\tan A + \sec A - 1}{\tan A - \sec A + 1} = \dfrac{1 + \sin A}{\cos A}$

$(\sec^2 A - \tan^2 A)$

$(\sec A + \tan A)\left(1 - (\sec A - \tan A)\right)$

DIBY-06

Prove that: $\sin^8 A - \cos^8 A = (\sin^2 A - \cos^2 A)(1 - 2\sin^2 A\cos^2 A)$

DIBY-07 (Hw)

$(1 + \tan\alpha\tan\beta)^2 + (\tan\alpha - \tan\beta)^2 =$

(A) $\tan^2\alpha\tan^2\beta$

(B) $\sec^2\alpha\sec^2\beta$

(C) $\tan^2\alpha\cot^2\beta$

(D) $\sec^2\alpha\cos^2\beta$

[Ans. : (B)]

DIBY-08

Prove that : $\frac{\sin A}{1+\cos A}+\frac{1+\cos A}{\sin A}=2\operatorname{cosec} A$

DIBY-09

Prove that : $\sqrt{\frac{1-\sin A}{1+\sin A}}=\sec A-\tan A$

DIBY-10

Prove that : $\frac{\operatorname{cosec} A}{\operatorname{cosec} A-1}+\frac{\operatorname{cosec} A}{\operatorname{cosec} A+1}=2\sec^2 A$

DIBY-11

Prove that : $\frac{\operatorname{cosec} A}{\cot A+\tan A}=\cos A$



Homework

Re-attempt all Class Examples

DPP & DIBY

Module: Only 2 Exercise

Topicwise & Bridge(Setu)- 4 sittings each of 1 HR

It is never about end result

It's always about Journey

Thank You

#futureIITians